

We should probably start with just RUP then RUP+UR and then try some other small combinations. $\text{RAT}_\forall$ is probably the last rule we need to verify.

Crnv	RUP	$\text{RAT}_\exists$	Uadd	Eadd	Dadd	Drmv	Ured	$\text{RAT}_\forall$	compl	pr.cmp.
	✓								UnSat	Resolution
	✓	✓							UnSat	RAT^-
✓	✓	✓							UnSat	DRAT^-
	✓	✓		✓					UnSat	RAT
✓	✓	✓		✓					UnSat	DRAT
	✓						✓		FQBF	QU-Res
	✓					✓	✓		FQBF	QU(D)-Res
	✓	✓		✓	✓		✓		FQBF	Ext QU-Res
✓	✓	✓	✓	✓	✓				TDQBF	
	✓	✓	✓	✓	✓	✓	✓		FDQBF	IndExtQU-Res
✓	✓	✓	✓	✓	✓	✓	✓		DQBF	IndExtQU-Res
✓	✓	✓	✓	✓	✓	✓	✓	✓	DQBF	IndExtQU-Res

Fig. 1. Combinations of sound rules $\text{DQRAT} + \text{D}^{\forall \text{pure}}$. Complete for different problems, and p-equivalent to known proof systems.

Let $x^0 = \neg x$ and $x^1 = x$. Note this should not be confused with the c++ code where x^1 , is $\neg x$ where '^' is the bitwise XOR.

Lemma 1 (Clause Removal). *Suppose we have a DQBF $(\Pi\phi)$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E and contains at least one clause.*

Then if $\Pi\phi$ is satisfiable, for any clause $C \in \phi$, $\Pi\phi \setminus \{C\}$ is satisfiable.

Proof. Suppose $\Pi\phi$ is satisfiable, i.e. there is a set of Skolem functions $f = \{f_e \mid e \in E\}$ (with appropriate domains i.e. $\text{dom}(f_e) = D_e$) that satisfy all clauses of ϕ under every universal assignment.

Suppose we have some universal assignment α we can extend it to $\beta = \alpha \cup \{e^{f_e(\alpha|_{D_e})} \mid e \in E\}$ and it will satisfy ϕ , i.e.e satisfy all clauses in ϕ . removing a clause in ϕ still means every remaining clause is satisfied under this assignment. So $\phi \setminus \{C\}$ by this extended assignment, and we can generalise this to all universal assignments and their extensions. Therefore $\Pi\phi \setminus \{C\}$ is satisfiable.

Suppose we have some clause C then its negation $\bar{C}$ is written as a conjunction of singleton clauses $\bigwedge_{c \in C} \neg c$.

Lemma 2. *Suppose we have a DQBF $(\Pi\phi|_\alpha)$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E . α is a Boolean assignment to the variables of ϕ .*

In addition suppose we have an array of literals C none satisfied by α , and no pair of literals is complementary.

Then we apply **negate_and_propagate**(C) to the state $(\Pi\phi|_\alpha)$ and return false, then the resulting state $(\Pi\phi|_\beta)$ is such that the propositional formulas ϕ_β and $\phi|_{\alpha \cup \{\neg c | c \in C\}}$ are logically equivalent, in other words they have exactly the same models.

Proof. **negate_and_propagate** begins at α and queues both literals from $\{\neg c | c \in C\}$ and any watch literals that become unit in the process. It terminates at a conflict to return true i.e. when β contradicts with a literal l on the propagation queue or when the queue is empty and all variables from $\{\neg c | c \in C\}$ have already been added and it returns false. The assignment is monotonic therefore if it returns false $\beta \supseteq \alpha \cup \{\neg c | c \in C\}$ and so $\phi_\beta \Rightarrow \phi_{\alpha \cup \{\neg c | c \in C\}}$.

So at any stage if the assignment is γ and a literal l is added from a unit clause it must have been implied i.e. $\phi_\gamma \Rightarrow l$. Later when it is at the front of the propagation queue we may have a larger assignment δ but since $\gamma \subseteq \delta$, then $\phi_\delta \Rightarrow \phi_\gamma$ and so $\phi_\delta \Rightarrow \phi_{\delta \cup \{l\}}$. This means via induction and monotonicity that $\phi_{\alpha \cup \{\neg c | c \in C\}} \Rightarrow \phi_\beta$.

Lemma 3. Suppose we have a DQBF $(\Pi\phi|_\alpha)$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E . α is a Boolean assignment to the variables of ϕ .

In addition suppose we have an array of literals C none satisfied by α .

Then we apply **negate_and_propagate**(C) to the state $(\Pi\phi|_\alpha)$. If **conflict==true**, i.e. the process returns true, then $\phi_\alpha \wedge \bigwedge \{\neg c | c \in C\}$ is propositionally inconsistent.

Proof. As argued in the previous proof. At any stage if the assignment is γ and a literal l is added from a unit clause it must have been implied i.e. $\phi_\gamma \Rightarrow l$. Later when it is at the front of the propagation queue, we may have a larger assignment δ but since $\gamma \subseteq \delta$, then $\phi_\delta \Rightarrow \phi_\gamma$ and so $\phi_\delta \Rightarrow \phi_{\delta \cup \{l\}}$.

This means if l is added to the propagation queue via unit propagation at assignment γ then if the final assignment β conflicts with l , ϕ_β falsifies a clause. By monotonicity $\phi_\alpha \wedge \bigwedge \{\neg c | c \in C\}$ logically implies β so is inconsistent. Likewise if l is added because $\neg l$ is in C but $\neg l$ also appeared in β then $\phi_\alpha \wedge \bigwedge \{\neg c | c \in C\}$ logically implies both l and $\neg l$. Hence inconsistent. Finally we could have that C contains a complementary pair in which case we know $\phi_\alpha \wedge \bigwedge \{\neg c | c \in C\}$ is propositionally inconsistent. These exhaust all ways that we can return true.

Corollary 1 (RUP). Suppose we have a DQBF $(\Pi\phi)$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E . In addition suppose we have a clause C such that **negate_and_propagate**(C) at state $(\Pi\phi|_\alpha)$ returns true (i.e. there is a conflict).

Then if $\Pi\phi$ is satisfiable then so is $\Pi\phi \wedge C$

Proof. Suppose $\Pi\phi$ is satisfiable, i.e. there is a set of Skolem functions $f = \{f_e \mid e \in E\}$ (with appropriate domains i.e. $\text{dom}(f_e) = D_e$) that satisfy all clauses of ϕ under every universal assignment.

Suppose we have some universal assignment α we can extend it to $\beta = \alpha \cup \{e^{f_e(\alpha|_{D_e})} \mid e \in E\}$ and it will satisfy ϕ . We argue that it will also satisfy C . Suppose it does not, then it falsifies all literals in C . If C contains a complementary pair of literals it will be satisfied by any total assignment, so we know this cannot only happen when this is not the case. Therefore $\{\neg c \mid c \in C\}$ is a consistent partial assignment. But since **negate_and_propagate**(C) returns conflict, by the previous lemma we know $\phi_{\{\neg c \mid c \in C\}}$ is a contradiction, hence β that extends it cannot have satisfied ϕ to begin with

Theorem 1 (DQRATE). *Suppose we have a DQBF ($\Pi\phi$) with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E .*

*Suppose C is clause containing the existential literal l . And now suppose for every clause D with $\bar{l} \in D$ that **negate_and_propagate**(O_D^l) returns a conflict (true) at state $\Pi\phi_{\{\neg c \mid c \in C\}}$. O_D^l is the outer clause (wrt l) of D , it is a subset of $D \setminus \{\bar{l}\}$ that contains only universal literals that of universals in $x = \text{var}(l)$'s dependency set and existential literals whose dependency set is a subset x 's.*

Then if $\Pi\phi$ is satisfiable then so is $\Pi\phi \wedge C$

Proof. The argument works from Skolem modification. Suppose $\Pi\phi$ is satisfiable, i.e. there is a set of Skolem functions $f = \{f_e \mid e \in E\}$ (with appropriate domains i.e. $\text{dom}(f_e) = D_e$) that satisfy all clauses of ϕ under every universal assignment. We will modify f_x to f'_x so that ϕ remains satisfied under all assignments.

If $x = l$ then we define f'_x as true if f_x is true or if $\bigwedge_{\bar{l} \in D} O_D^l$ is true (i.e all outer clauses of D (wrt l) such that $\bar{l}$ is in D are simultaneously satisfied) and false otherwise. If $x = \neg l$ then we define f'_x as false if f_x is false or if $\bigwedge_{\bar{l} \in D} O_D^l$ is true, and true otherwise.

The outer condition is critical because the domain of f_x restricts it to just universal variables in D_x , we can also synchronise it with other Skolem functions f_e whose results would be known from this domain (exactly those existential variables e in the outer set of l). I.e. O_D^l is completely determined by D_x .

Let $x^0 = \neg x$ and $x^1 = x$. Suppose we have universal assignment α , we know that extending with f satisfies ϕ . i.e $\beta = \alpha \cup \{e^{f_e(\alpha|_{D_e})} \mid e \in E\}$ satisfies ϕ . Now we argue that $\beta' = \alpha \cup \{x^{f'_x(\alpha)}\} \cup \{e^{f_e(\alpha|_{D_e})} \mid e \in E \setminus \{x\}\}$ satisfies $\phi \wedge C$.

Suppose $\alpha \cup \{e^{f_e(\alpha|_{D_e})} \mid e \in E \setminus \{x\}\}$ satisfies $\bigwedge_{\bar{l} \in D} O_D^l$ then f'_x satisfies l and satisfies C , and any clauses that contain l in ϕ . For clauses that contain $\bar{l}$ they are already satisfied because $\bigwedge_{\bar{l} \in D} O_D^l$ is true. The remaining clauses that contain no l literal remain satisfied by $\alpha \cup \{e^{f_e(\alpha|_{D_e})} \mid e \in E \setminus \{x\}\}$ as this part is unchanged from β .

Now suppose $\alpha \cup \{e^{f_e(\alpha|_{D_e})} | e \in E \setminus \{x\}\}$ falsifies $\bigwedge_{D \in \phi}^{l \in D} O_D^l$, then it will behave exactly as β and satisfy ϕ . It must also satisfy some literal in C otherwise ϕ cannot have been true because $\phi_{\{\neg c | c \in C\}} \wedge \neg O_D^l$ is a contradiction for some D from the previous lemma.

Note that if a clause is RUP it satisfies the DQRATE property (as long as it has an existential variable).

Lemma 4 (Universal Addition). *Suppose we have a DQBF $\Pi\phi$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E . Let x be a variable not appearing in Π .*

Then if $\Pi\phi$ is satisfiable so is $\forall x\Pi\phi$, where no dependency set is altered.

Proof. Let $f = \{f_e \mid e \in E\}$ be a set of Skolem functions that satisfy $\Pi\phi$, in other words for any universal assignment α to U , then $\beta = \alpha \cup \{e^{f_e(\alpha|_{D_e})} | e \in E\}$ satisfies ϕ .

Now suppose we have some complete assignment γ to $U \cup \{u\}$, there is some assignment α to U and some assignment l to $\{u\}$ such that $\gamma = \alpha \cup \{l\}$. Now, let $\delta = \gamma \cup \{e^{f_e(\gamma|_{D_e})} | e \in E\}$, then $\delta = \alpha \cup \{l\} \cup \{e^{f_e(\alpha|_{D_e})} | e \in E\}$ and since β satisfies a literal in every clause δ simply extends it. So δ satisfies every clause in ϕ . Since this works with every α $\forall x\Pi\phi$ is satisfiable.

Lemma 5 (Existential Addition). *Suppose we have a DQBF $\Pi\phi$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E . Let x be a variable not appearing in Π .*

Then if $\Pi\phi$ is satisfiable so is $\Pi\exists x(\emptyset)\phi$, where no other existing dependency set is altered.

Proof. Let $f = \{f_e \mid e \in E\}$ be a set of Skolem functions that satisfy $\Pi\phi$, in other words for any universal assignment α to U , then $\beta = \alpha \cup \{e^{f_e(\alpha|_{D_e})} | e \in E\}$ satisfies ϕ . Let f_x be the constant function that maps to 0. $\gamma = \alpha \cup \{e^{f_e(\alpha|_{D_e})} | e \in E \cup \{x\}\}$ satisfies ϕ , because β is a sub-assignment and satisfies a literal in every clause. Therefore generalising to all α , $\Pi\exists x(\emptyset)\phi$ is satisfiable.

Lemma 6 (Dependency Addition). *Suppose we have a DQBF $\Pi\exists x(D_x)\phi$. Π contains a list of quantified universal variables U and a list of existential variables E with Henkin quantifier each existential variable e is paired with a dependency set D_e a subset of U . D_x is the dependency set of x and is also subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E and x .*

Suppose there is some $u \in U$ not appearing to D_x . Then if $\Pi\exists x(D_x)\phi$ is satisfiable then so is $\Pi\exists x(D_x \cup \{u\})\phi$.

Proof. Let $f = \{f_e \mid e \in E' \cup \{x\}\}$ be a set of Skolem functions that satisfy $\Pi\exists x(D_x)\phi$. We will construct the set of Skolem functions $f' = \{f'_e \mid e \in E \cup \{x\}\}$ that satisfy $\Pi\exists x(D_x)\phi$. For $e \neq x$ let $f'_e = f_e$. Then for f'_x we to deal with a larger domain. Let α be an assignment to D_x then define $f'_x(\alpha \cup \{u\}) = f_x(\alpha) = f_x(\alpha \cup \{\bar{u}\})$. f' will satisfy ϕ because when we have some assignment γ to U and we extend it we get $\beta = \gamma \cup \{e^{f_e(\gamma|_{D_e})} \mid e \in E\} \cup \{x^{f_x(\gamma|_{D_x})}\} = \gamma \cup \{e^{f_e(\gamma|_{D_e})} \mid e \in E\} \cup \{x^{f_x(\gamma|_{D_x \cup u})}\}$ which satisfies every clause in ϕ .

For a DQBF $\Pi\phi$, we will denote membership $(u, x) \in \mathcal{D}^{\forall\text{pure}}(\Pi\phi)$ to mean that $\exists$ -variable x really does depend on $\forall$ variable u after considering the pure universal dependency scheme.

Definition 1. Let ψ be a DQBF, χ a subset of the clauses in ψ , $\mathcal{S}$ a subset of variables appearing in ψ and u a universal literal. We define the sets $\mathcal{C}^{\forall\text{pure}}(u, \psi, \chi, \mathcal{S})$ and $\mathcal{L}^{\forall\text{pure}}(u, \psi, \chi, \mathcal{S})$. Initially $\mathcal{C}^{\forall\text{pure}}(u, \psi, \chi, \mathcal{S})$ contains all clauses from χ and $\mathcal{L}^{\forall\text{pure}}(u, \psi, \chi, \mathcal{S})$ contains all $\mathcal{S}$ -literals in χ . We expand these sets in a similar way as before, suppose $p \in \mathcal{L}^{\forall\text{pure}}(u, \psi, \chi, \mathcal{S})$ we include any ψ clause E with $\bar{p} \in E$ and **crucially** $\bar{u} \notin E$ into $\mathcal{C}^{\forall\text{pure}}(u, \psi, \chi, \mathcal{S})$, and also include the literals $\{x \in E \mid x \neq \bar{p}, x \in \mathcal{S}\}$ into $\mathcal{L}^{\forall\text{pure}}(u, \psi, \chi, \mathcal{S})$, and as we increase this set we can further propagate until we reach fix-point.

Definition 2. Given a DQBF $\Pi\phi$ with CNF matrix ϕ . We say there is a pure path from universal literal u to existential literal x (denoted $u \sim_{\forall\text{pure}} x$) if and only if $x \in \mathcal{L}^{\forall\text{pure}}(u, \psi, \phi_u, \mathcal{S})$. Where ϕ_u is the set of clauses that contain literal u and $\mathcal{S}$ is the set of $\exists$ variables that contain u in its dependency set.

Definition 3. Given a DQBF $\Pi\phi$ with CNF matrix ϕ , let (u, x) be a pair with a $\forall$ variable u and $\exists$ variable x . $(u, x) \in \mathcal{D}^{\forall\text{pure}}(\Pi\phi)$ ($\in \mathcal{D}^{\forall\text{pure}}$ when unambiguous) if and only if $u \in D_x$ and either

1. $u \sim_{\forall\text{pure}} x$ and $\bar{u} \sim_{\forall\text{pure}} \bar{x}$, or
2. $\bar{u} \sim_{\forall\text{pure}} x$ and $u \sim_{\forall\text{pure}} \bar{x}$.

Lemma 7 (Dependency Removal). Suppose we have a DQBF $\Pi\exists x(D_x)\phi$. Π contains a list of quantified universal variables U and a list of existential variables E with Henkin quantifiers, each existential variable e is paired with a dependency set D_e a subset of U . D_x is the dependency set of x and is also subset of U . ϕ is the propositional matrix written as a conjunctive normal form using only variables from U and E and x .

Suppose there is some $u \in D_x$, and none of these conditions are satisfied

- a u -pure path from u to l and a $\bar{u}$ -pure path from $\bar{u}$ to $\bar{l}$
- a u -pure path from u to $\bar{l}$ and a $\bar{u}$ -path from $\bar{u}$ to l

Then if $\Pi\exists x(D_x)\phi$ is satisfiable then so is $\Pi\exists x(D_x \setminus \{u\})\phi$.

Proof. Let $f = \{f_e \mid e \in E\}$ be a set of Skolem functions that satisfies $\Pi\exists x(D_x)\phi$. We will construct a set of Skolem functions $f^* = \{f_e^* \mid e \in E\}$ that satisfies $\Pi\exists x(D_x \setminus \{u\})\phi$.

We use the following definitions:

- If α is partial assignment to the universal variables, α^v is obtained by flipping the Boolean value of universal variable v .
- Given f a *dependency witness* is a triple (α, y, v) where $f_y(\alpha) \neq f_y(\alpha^v)$, y is an existential variable, v is a universal variable, α is an assignment to D_y
- The set $\mathcal{S}_u = \{z : u \in D_z\}$
- For universal literal $l: \phi_l = \{C \in \phi \mid l \in C\}$.
- For an existential variable z , and assignment $\gamma \in \{0, 1\}^U$: $\gamma_z = \gamma|_{D_z}$

Suppose f has a dependency witness (α, x, u) where $u \in D_x^{\Pi} \setminus D_x^{\Pi'}$. Let l_u be the literal on u satisfied by α , l_x the literal on x satisfied by $f_x(\alpha)$. Since $u \notin D_x^{\Pi'}$, by definition either $\bar{l}_u \not\sim_{\forall\text{pure}} l_x$ or $l_u \not\sim_{\forall\text{pure}} \bar{l}_x$. Without loss of generality, let $\bar{l}_u \not\sim_{\forall\text{pure}} l_x$ (in the other case, swap the roles of α and α^u).

Define $\mathcal{M} = \mathcal{M}(f, \bar{l}_u, l_x, \alpha)$ as the set of all Skolem sets $f' = \{f'_x \mid x \in E\}$ respecting the prefix Π with the following properties:

1. The set of dependency witnesses in f' and u are a *proper* subset of dependency witness in f and u . i.e if f' has a dependency witness (δ, y, u) then f has dependency witness (δ, y, u) and there is least one dependency witness for f that is not present for f' .
2. For every existential variable z , for every $\gamma \in \{0, 1\}^{D(z)}$, if $f'_z(\gamma_z) \neq f_z(\gamma_z)$ then γ_z satisfies l_u ; i.e. $u \in D(z)$ and l_u 's polarity matches with γ_z .
3. For every existential variable z , for every $\gamma \in \{0, 1\}^{D(z)}$, if $f'_z(\gamma_z) \neq f_z(\gamma_z)$ then for the literal l_z satisfied by $f_z(\gamma_z)$, $\bar{l}_u \not\sim_{\forall\text{pure}} l_z$.

We first show that $\mathcal{M}$ is non-empty. Let

$$f^0 = \{f_z^0 \mid z \in E\} \quad f_z^0(\tau) = \begin{cases} \neg f_x(\tau) & \text{if } \tau = \alpha \text{ and } z = x \\ f_z(\tau) & \text{otherwise,} \end{cases}$$

1. f^0 satisfies property 1 because (α, x, u) and (α^u, x, u) are dependency witnesses that are removed. For any dependency witness (δ, w, u) present in f^0 then $w \neq x$ or $(\delta \neq \alpha \text{ and } \delta \neq \alpha^u)$ and so $f_w^0(\delta) = f_w(\delta)$ and $f_w^0(\delta^u) = f_w(\delta^u)$ so (δ, w, u) is a dependency witness in f .
2. f^0 only differs from f on α and its extensions, but α satisfies l_u .
3. f_z^0 only differs from f_z when $z = x$ however we have assumed that $\bar{l}_u \not\sim_{\forall\text{pure}} l_x$ and $f_x(\alpha)$ satisfies l_x .

Next we will define the inductive step. We show that if $f' \in \mathcal{M}$ is not a model then there is $f'' \in \mathcal{M}$ where the dependency witnesses of f'' and u are a proper subset of the dependency witnesses of f' .

First we need to construct such a f'' . We know that if f' is not a model of $\Pi \exists x(D_x)\phi$, then there is some assignment $\gamma \in \{0, 1\}^E$ and some clause $C \in \phi$ such that the assignment $\gamma \cup f'(\gamma)$ falsifies C . However we know that $\gamma \cup f(\gamma)$ satisfies C because f is a model. Hence there is some existential variable z such that $f'_z(\gamma_z) \neq f_z(\gamma_z)$ and some literal l_z in z such that $f_z(\gamma_z)$ satisfies l_z and $f'_z(\gamma_z)$ falsifies l_z . By property 2: γ_z satisfies l_u , and by property 3: $\bar{l}_u \not\sim_{\forall\text{pure}} l_z$. On γ^u , we have that l_u is falsified so f' and f are equal by property 2. This

means that $f'_z(\gamma_z^u) = f_z(\gamma_z^u)$. Therefore if $f_z(\gamma_z) \neq f_z(\gamma_z^u)$ (i.e. (γ_z, z, u) is not a dependency witness in f) then $f'_z(\gamma_z) \neq f_z(\gamma_z) = f_z(\gamma_z^u) = f'_z(\gamma_z^u)$ (i.e. (γ_z, z, u) is a dependency witness in f') violating property 1, so $f_z(\gamma_z^u) \neq f_z(\gamma_z^u)$ and $f'_z(\gamma_z) = f'_z(\gamma_z^u)$.

This means there is some other literal $l_y \in C$ with variable y such that it is satisfied in $\gamma^u \cup f(\gamma^u)$. As we observed that f and f' must agree on γ^u , l_y is also satisfied by $\gamma^u \cup f'(\gamma^u)$. $\gamma \cup f(\gamma)$ does not satisfy l_y as it does not satisfy clause C . Therefore the only universal variable y can be is u , however l_y cannot be l_u , otherwise $\gamma \cup f(\gamma)$ would satisfy this (we come back to this for pure paths), and l_y cannot be $\bar{l}_u$ otherwise we would have $\bar{l}_u \not\sim_{\forall \text{pure}} l_z$. Therefore y is existential. Let

$$f'' = \{f''_w \mid w \in E\} \quad f''_w(\tau) = \begin{cases} \neg f'_w(\tau) & \text{if } \tau = \gamma_y \text{ and } w = y \\ f'_w(\tau) & \text{otherwise,} \end{cases}$$

we show that $f'' \in \mathcal{M}$:

1. (+ smaller than f') As $\gamma \cup f(\gamma)$ falsifies l_y and $\gamma \cup f(\gamma^u)$ satisfies l_y , (γ_y, y, u) and (γ_y^u, y, u) are dependency witnesses for f' and therefore must be for f by induction hypothesis. $f''_y(\gamma_y) = f''_y(\gamma_y^u)$ and thus these dependency witnesses are removed for f'' . For any dependency witness (δ, w, u) in f'' , $w \neq y$ or $(\delta \neq \gamma_y \text{ and } \delta \neq \gamma_y^u)$, and so $f''_w(\delta) = f'_w(\delta)$ and $f''_w(\delta^u) = f'_w(\delta^u)$ so (δ, w, u) is a dependency witness in f' and thus also in f .
2. f'' only differs from f' on γ_y and its extensions. Since (γ_y, y, u) is a dependency witness for f' , then $u \in D(y)$, and since it is consistent with γ then γ_y satisfies l_u .
3. f''_w only differs from f'_w when $w = y$. We know that (γ_y, y, u) is a dependency witness for f , so as $f_y(\gamma_y^u)$ satisfies l_y then $f_y(\gamma_y)$ satisfies $\bar{l}_y$. Assume for contradiction, that $\bar{l}_u \sim_{\forall \text{pure}} \bar{l}_y$. We know that if l_u was in C then γ would satisfy C which it does not (this is the only observation needed to get from $\mathcal{D}^{\text{rrs}}$ to $\mathcal{D}^{\forall \text{pure}}$). Hence extending $\bar{l}_u \sim_{\forall \text{pure}} \bar{l}_y$ through the positive l_y to l_z gives us a $\bar{l}_u$ pure path from $\bar{l}_u$ to l_z giving us a contradiction.

Since the set of dependency witnesses is finite the induction step eventually terminates with model where (α, x, u) is no longer a dependency witness, and we have strictly fewer dependency witnesses on u . We can repeat this for any other witness (α', x, u) until all witnesses of (α', x, u) are removed to get model f^* . We can restrict the domain of f^*_x to remove u and it will still be well defined and a winning Skolem function.

Lemma 8 ($\forall$ red soundness).

Suppose we have a DQBF $\Pi\phi \wedge (C \vee l)$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . l is a universal literal, C is a clause and ϕ is a formula in conjunctive normal form. $\phi \wedge (C \vee l)$ is the propositional matrix and uses only variables from U and E . Suppose $\neg l$ not in C and C contains no existential variables x such that u is in the dependency set of x . Then if $\Pi\phi \wedge (C \vee l)$ is satisfiable then $\Pi\phi \wedge (C \vee l) \wedge C$ is satisfiable.

Proof. An S-form DQBF is true if and only if it has a set f of satisfying Skolem functions, a function f_e for each of its existential variables e . Suppose $\Pi\phi \wedge (C \vee l)$ is satisfied by the set $f = \{f_x \mid x \in E\}$. We will show that $\Pi\phi \wedge (C \vee l) \wedge C$ is satisfied by the same functions.

Consider an arbitrary universal assignment α and we extend it with $\beta = \alpha \cup \{e^{f_e(\alpha|_{D_e})} \mid e \in E\}$, $\phi \wedge (C \vee l)$ is satisfied by β , the question is whether C is also satisfied. We are assuming that clauses do not contain repeated literals so l is not in C . If α falsifies l then C must be satisfied otherwise C could not be satisfied. If α satisfies l then $\alpha \setminus \{l\}$ completely determines C because $\text{var}(l)$ is not in the dependency set of any existential literals in C and $\neg l$ not in C . This means α and $(\alpha \setminus \{l\}) \cup \{\neg l\}$ must determine C in the same way under f and we know $(\alpha \setminus \{l\}) \cup \{\neg l\}$ when extended must satisfy C hence β satisfies C .

Lemma 9 (DQRATU soundness).

Suppose we have a DQBF $\Pi\phi \wedge (C \vee l)$ with quantifier prefix Π containing a list of universal variables U and a list of existential variables E each existential variable e is paired with a dependency set D_e a subset of U . l is a universal literal, C is a clause and ϕ is a formula in conjunctive normal form. $\phi \wedge (C \vee l)$ is the propositional matrix and uses only variables from U and E .

And now suppose for every clause $D \in \phi$ with $\bar{l} \in D$ that either:

1. There is no $\bar{l}$ -pure path in the variables that depend on $\text{var}(l)$ from D to C in $\Pi\phi \wedge C$.
2. **negate_and_propagate**(O_D^l) returns a conflict (true) at state $\Pi\phi_{\{\neg c \mid c \in C\}}$. O_D^l is the outer clause (wrt l) of D , it is a subset of $D \setminus \{\bar{l}\}$ that contains only universal literals that of universals in $x = \text{var}(l)$'s dependency set and existential literals whose dependency set is a subset x 's.

Then if $\Pi\phi \wedge (C \vee l)$ is satisfiable then $\Pi\phi \wedge (C \vee l) \wedge C$ is satisfiable.

Proof. Let $f = \{f_e \mid e \in E\}$ be a set of Skolem functions that satisfies $\Pi\phi \wedge (C \vee l)$. We will construct a set of Skolem functions $f^* = \{f_e^* \mid e \in E\}$ that satisfies $\Pi\phi \wedge C$.

Let the following set be defined

- $\mathcal{S} = \{z : \text{var}(l) \in D_z\}$
- $\mathfrak{L}^{\forall\text{pure}}(\bar{l}_u, \phi \wedge, \{C\}, \mathcal{S})$ is the set of exit literals on $\bar{l}_u$ -pure resolution paths in variables of $\mathcal{S}$, starting from $\{C\}$ ranging over $\phi \wedge C$.
- $\mathfrak{C}^{\forall\text{pure}}(\bar{l}_u, \phi \wedge, \{C\}, \mathcal{S})$ is the set of clauses on $\bar{l}_u$ -pure resolution paths in variables of $\mathcal{S}$, starting from $\{C\}$ ranging over $\phi \wedge C$.
- $\chi = \{D \in \phi \mid \neg l \in D\} \cap \mathfrak{C}^{\forall\text{pure}}(\bar{l}_u, \phi \wedge C, \{C\}, \mathcal{S})$

We consider $\mathfrak{L}^{\forall\text{pure}}(\bar{l}_u, \phi \wedge, \{C\}, \mathcal{S})$ (the set of exit literals on $\bar{l}_u$ -pure resolution paths in variables of $\mathcal{S}$, starting from $\{C\}$ ranging over $\phi \wedge C$). And their variables will be those that we modify.

$$f_x^*(\alpha) = \begin{cases} f_x(\alpha \setminus \{l\} \cup \{\bar{l}\}) & \bigwedge_{D \in \chi} O_D \text{ is satisfied under } \alpha \\ & \exists l \in \mathfrak{L}^{\forall\text{pure}}(\bar{l}_u, \phi \wedge C, \{C\}, \mathcal{S}) \text{ s.t. } \text{var}(l) = x, \\ f_x(\alpha) & \text{otherwise.} \end{cases}$$

We now need to argue that under any universal assignment γ these modified functions will satisfy every clause in ϕ as well as C . First suppose $\bigwedge_{D \in \chi} O_D$ is satisfied under γ . Then the existential variables in C , act according $f_x(\gamma|_{D_x} \setminus \{l\} \cup \{\bar{l}\})$ which must satisfy the clause, because u does not under $\gamma \setminus \{l\} \cup \{\bar{l}\}$. We also know any clause that has an l literal in must be satisfied because if γ includes l then it is satisfied through that literal or if not the functions are all unchanged and retain satisfaction. Now clauses that are connected to C via a pure path without an l or $\bar{l}$ literal, have all their $\mathcal{S}$ literals synchronised so either they play as γ or $\gamma \setminus \{l\} \cup \{\bar{l}\}$ but must satisfy in the original case and so must again. Clauses not connected via a pure path retain the same skolems functions on all variables. The remaining clauses are those in χ but these already are satisfied because their outer clauses are.

Next if some outer clause from χ is falsified, we know that propositionally C is implied and since the Skolem functions remain the same then all other clauses are satisfied.